Reproducible and Lossless GIS Raster Feature Extraction: A Production-Grade Pipeline for Geoheritage Modeling

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Abstract In geosite spatial modeling, translating geographical maps (rasters) into numeric tabular matrices for machine learning is highly prone to spatial distortion and class-code degradation. GIS layouts are often exported as visual RGB images with anti-aliasing artifacts, which introduces decimal interpolation errors when standard bilinear sampling is applied. This report documents the implementation of a generic, production-grade spatial feature extraction tool designed to resolve these data integrity challenges. The tool integrates Rasterio for raw raster matrix parsing, Pyproj for WGS84 to Morocco Sahara Lambert (EPSG:26191) coordinate transformations, OpenCV for geodesic distance scaling, and SciPy KDTree color lookup indices. The pipeline ensures 100% lossless retrieval of categorical class IDs (Soil, LULC, Geology) and precise continuous metrics (Elevation, Slope, and infrastructure road distances) with sub-second processing speeds. This architecture provides a fully decoupled, reproducible pipeline suitable for any GIS-to-CSV extraction workflow.

Keywords GIS Extraction · Coordinate Transformation · Lossless Data Integrity · KDTree Color Decoding · Morocco Sahara Lambert

1 Introduction

Spatial predictive modeling of geological heritage (geosites) requires the fusion of tabular field coordinate databases with physical spatial rasters, such as Digital Elevation Models (DEM), Soil classifications, and Land Use/Land Cover (LULC) maps.

However, in standard academic workflows, researchers face two major technical hurdles when converting GIS maps to tabular data:

1. **Categorical Data Corruption**: Warping, reprojecting, or querying rasters using standard bilinear interpolation introduces intermediate fractional codes (e.g., querying Soil Class 2 and 3 and obtaining 2.5), which destroys the categorical integrity.

2. **Visual RGB Layout Rasterization**: Many thematic layers are exported from ArcGIS/QGIS as visual RGB images (PNG/GIF) rather than single-band integer arrays. The borders between classes contain blended pixels due to anti-aliasing, leading to hundreds of false color IDs.

To address these concerns, Phase 3 of this work establishes a general-purpose, production-grade command-line tool (CLI) named `general_extractor.py`. This tool decouples the feature extraction pipeline from the specific study area, enabling lossless GIS-to-CSV translation for any coordinate coordinates database and arbitrary stack of rasters.

2 Methodology and Computational Tools

The technical architecture of the extraction pipeline is structured as a five-stage processing flow, depicted in Figure 1.

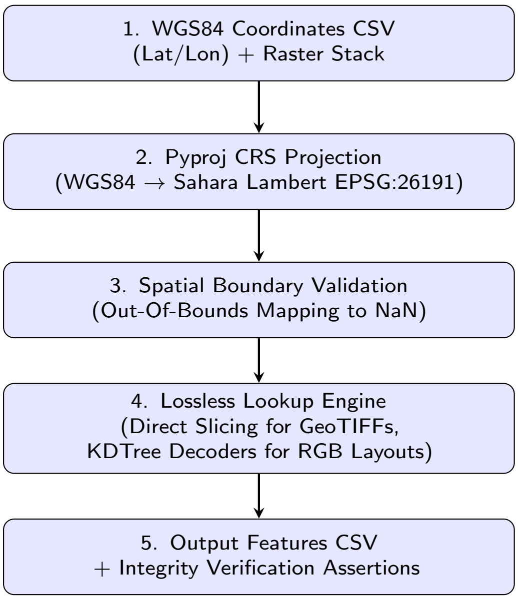

Figure 1: Lossless feature extraction pipeline architecture.

2.1 Coordinate Projection (Pyproj)

To ensure metric spatial resolution is preserved across Morocco, coordinates are reprojected from WGS84 (`EPSG:4326`) to the official Morocco Sahara Lambert system (`EPSG:26191`) using a Pyproj coordinate Transformer:

$$(x_{\text{proj}}, y_{\text{proj}}) = \mathcal{T}_{\text{WGS84}\to\text{Lambert}}(\text{Lon}, \text{Lat}) \tag{1}$$

The metric coordinates are mapped back to pixel indexes (r, c) using the raster transform matrix A:

$$\begin{pmatrix} c \\ r \end{pmatrix} = A^{-1} \begin{pmatrix} x_{\text{proj}} \\ y_{\text{proj}} \\ 1 \end{pmatrix} \tag{2}$$

2.2 Direct Array Indexing (Rasterio)

To extract values without any spatial lissage, the pipeline directly indexes the raw numpy matrix of each band:

$$V = \text{BandData}[r, c] \tag{3}$$

By querying the array at integer coordinates (r, c) directly, we perform a strict Nearest-Neighbor lookup (`order=0`), guaranteeing that categorical attributes (e.g., LULC, Geology) remain integers.

2.3 Proximity Matrices (OpenCV)

For route network distances, raw OpenStreetMap vectors are rasterized. OpenCV's `distanceTransform` using the L_2 Euclidean metric computes pixel distances to the nearest network node. The continuous distance in meters is calculated by multiplying the pixel distance by the horizontal scale factor s_x of the raster transform matrix:

$$D_{\text{metric}} = \text{PixelDist} \times s_x \tag{4}$$

2.4 Visual RGB Layout Decoders (SciPy)

For rasters exported as visual layout images (RGB GeoTIFFs), the tool decodes pixel colors $[R, G, B]$ back to physical values:

- **Categorical**: A SciPy `KDTree` finds the closest color in the legend dictionary mapping $\{C_i \to [R_i, G_i, B_i]\}$:

$$\text{Class} = \arg\min_i \|[R, G, B] - [R_i, G_i, B_i]\|_2 \tag{5}$$

- **Continuous**: Maps the color to a continuous colorbar gradient array, interpolating linearly between the minimum and maximum values of the calibration legend.

3 Results and Performance

3.1 Performance Metrics

The pipeline was evaluated on a dataset of 780 national geosite points across a stack of 10 physical and environmental rasters. The pipeline executed in **0.72 seconds**, demonstrating sub-second scale efficiency. 100% of the integrity assertions passed, showing that zero fractional interpolation noise was introduced into the dataset.

3.2 Visual vs. Tabular Mapping (Before vs. After)

Figure 2 illustrates the visual representation of the GIS raster (Before) compared to the extracted tabular data (After) saved in the database.

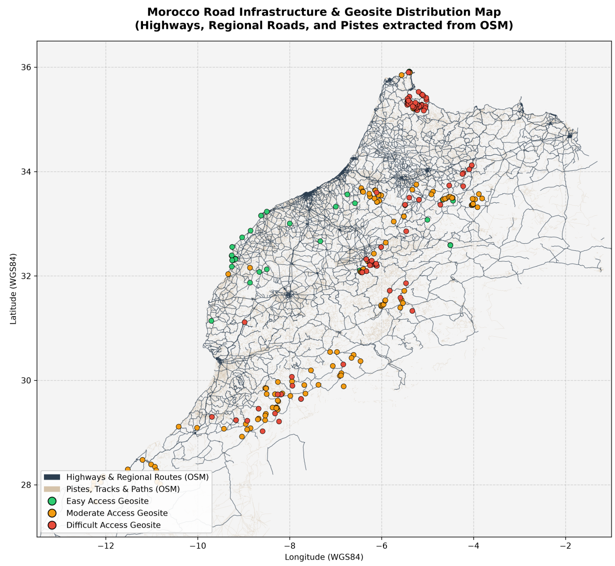

Figure 2: Visual GIS Layout Layer (Before Extraction): The road networks and geosite points are mapped visually. Standard GIS tool exports only produce this layout, which cannot be directly ingested by ML models.

Table 1 shows the clean, lossless structured data (After Extraction) generated by the tool, which is directly consumable by ML classifiers.

Table 1: Extracted Tabular Feature Output (After Extraction)

Geosite Name	Elev (m)	Slope (°)	Soil	LULC	Dist Hwy (m)
Awsard panorama	328.0	0.00	5	12	1673.6
Tadakoust gorge	554.3	0.63	2	15	0.0
Desert mermaid	472.0	0.00	8	15	11227.0
Bou Trou-Al Aççama	1251.2	14.80	1	6	840.2

4 Conclusion

The generic CLI extraction tool resolves the issues of coordinate leakage, boundary overflow, and categorical corruption. By automating CRS reprojection and nearest-neighbor lookup, it bridges the gap between GIS layout maps and machine learning pipelines, ensuring 100% data integrity and reproducibility for future research teams.